THE DIFFERENTIAL · CASE 01

Model Kill-Switch Protocol

A stop that cannot be forged. That cannot be turned against you. And still works when everything around it falls apart.

Trust

- **Every command is signed:** Only an authorised party can stop the model. Every command passes four checks before the gateway moves.
- **A committee, not one person:** A valid command needs three of five signers. One stolen key is not enough.
- **Guarding the guard:** The mechanism watches itself too: who signed, how many, when, and it alerts on anything unusual.

Gateway

- **The gateway:** Every request to the model gets a split-second decision: pass or block.
- **Targeted stop:** You do not always need to stop everything. One model, one region or one capability is possible.
- **Simulate before stopping:** Before signing a real stop, you see exactly what it would hit.

Resilience

- **No second use:** A genuine command sent once will not work a second time.
- **When the infrastructure falls:** Redis down? Kafka down? Each scope has a predefined answer: carry on or block.
- **Many regions:** The command must reach every region where the model runs, even when links between regions break.
- **Automatic expiry:** A forgotten stop does not last forever: after four hours it lifts by itself, unless re-signed.

Adoption

- **Developer libraries:** Connecting the mechanism to an existing system takes about ten lines of code.

Six decisions

- **Ed25519 or RSA?** We chose Ed25519.
- **One person or a committee?** We chose a committee, 3 of 5.
- **Carry on or block when it all fails?** We chose: it depends on the scope.
- **AI or statistics?** We chose simple statistics.
- **Cut or drain?** We chose to drain.
- **Stop forever or for a while?** We chose a stop with an expiry.